MATH49111 Scientific Computing
Analysis Exercise 1

The Logistic Map

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The Logistic Map

The logistic map is defined by

$$x_{n+1} = \alpha x_n(1 - x_n), \qquad x_0 = \frac{1}{2}.$$

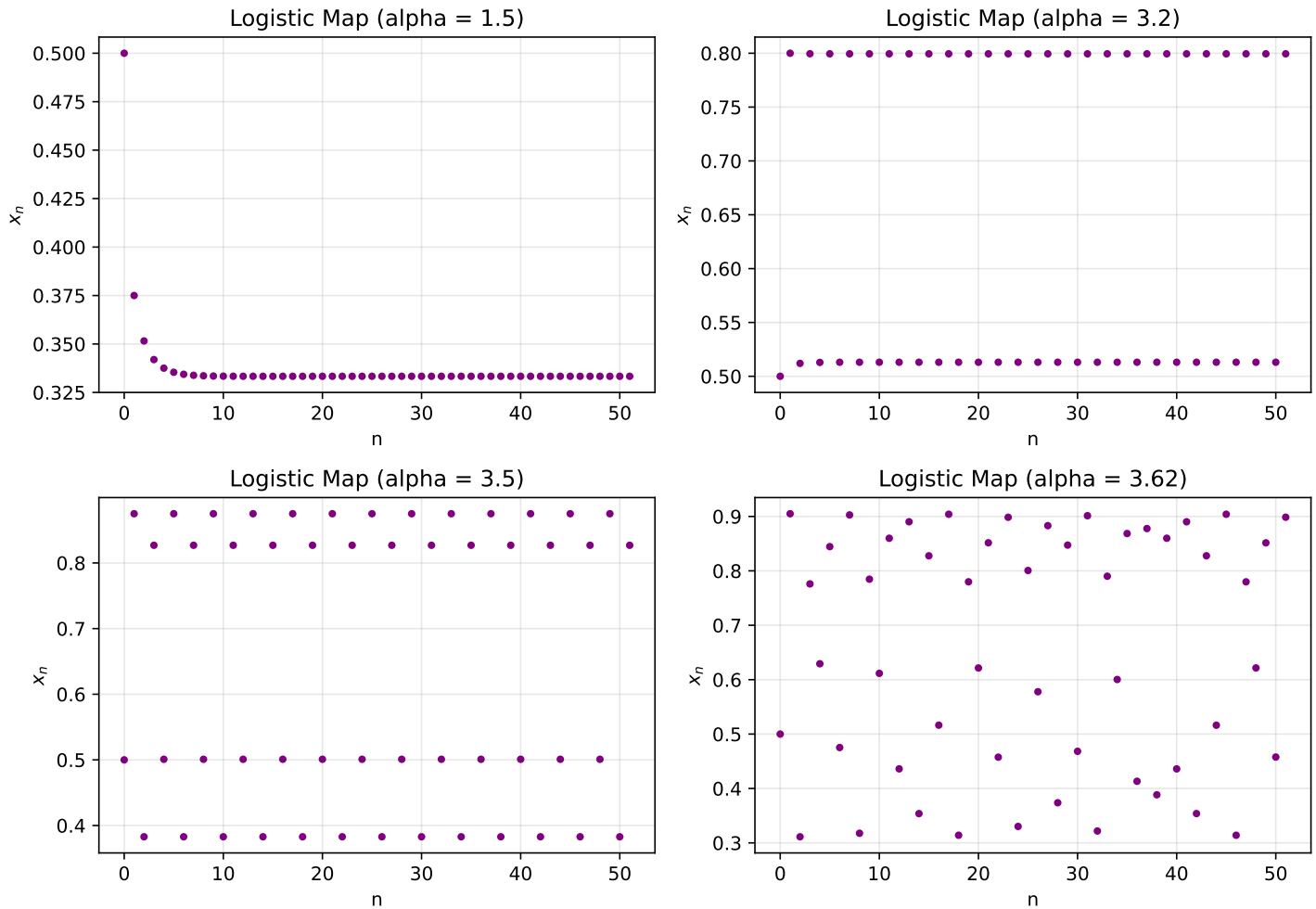
- Compute x_n for $n = 0, 1, \ldots, 50$.
- Investigate $\alpha = 1.5,\ 3.2,\ 3.5,\ 3.62$.

Observation

- $\alpha = 1.5$: converges to a fixed point around $1/3$.
- $\alpha = 3.2$: approaches a period-2 cycle.
- $\alpha = 3.5$: approaches a period-4 cycle.
- $\alpha = 3.62$: irregular behaviour: consistent with chaos.

Results: Iteration of the Logistic Map

Plots of x_n against n for each α.



Analytical Prediction for $\alpha = 1.5$

If the sequence converges to a fixed point x_∞, then

$$x_\infty = \alpha x_\infty (1 - x_\infty).$$

Rearranging

$$x_\infty (1 - \alpha + \alpha x_\infty) = 0.$$

Hence,

$$x_\infty = 0 \quad \text{or} \quad x_\infty = 1 - \frac{1}{\alpha}.$$

For $\alpha = 1.5$, the fixed point is

$$x_\infty = 1 - \frac{1}{1.5} = \frac{1}{3}.$$

The numerical results converge to 0.333..., agreeing with the analytical prediction.